

AN INTELLIGENT FRAMEWORK FOR RESUME INFORMATION EXTRACTION AND CANDIDATE-JOB MATCHING USING NLP AND SEMANTIC RETRIEVAL

AIMLCZG628T: DISSERTATION - MID-SEMESTER REPORT

by

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ABSTRACT

Recruitment teams process large volumes of resumes written in diverse formats and terminology. Traditional applicant tracking systems primarily depend on keyword search and rule-based filters, which often fail when equivalent skills, roles, and responsibilities are expressed differently. This dissertation develops an intelligent framework for resume information extraction and candidate-job matching using Natural Language Processing, structured information extraction, semantic retrieval, and ranking. The system accepts PDF, DOC, and DOCX resumes; validates and stores original files; extracts text using document parsers with an OCR fallback; and applies a hybrid resume parser to identify candidate details, skills, education, employment history, certifications, titles, companies, and experience. The extracted information is schema-validated, normalized into canonical candidate attributes, and indexed for structured, lexical, and semantic search. Job descriptions are similarly processed to identify mandatory and preferred skills, role, experience, education, certifications, responsibilities, domain, and location. Candidate retrieval combines keyword search, structured filters, and dense semantic similarity to produce a shortlist instead of scanning all source resumes. The shortlisted candidates are scored and reranked using skill coverage, title similarity, experience fit, domain relevance, and semantic evidence. Work completed up to the mid-semester stage includes literature review, system and data-model design, resume/JD preprocessing, baseline keyword and TF-IDF matching, resume extraction, JD requirement extraction, semantic representation, hybrid retrieval design, and an initial candidate-ranking pipeline. The remaining work focuses on controlled evaluation, ablation studies, recruiter-oriented explanations, scalability validation, and final dissertation reporting.

Keywords: Applicant Tracking System, Resume Parsing, Natural Language Processing, Information Extraction, Semantic Retrieval, Hybrid Search, Candidate Ranking, Sentence Embeddings.

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1. INTRODUCTION AND PROBLEM CONTEXT

The project addresses the technical core of an intelligent Applicant Tracking System: converting unstructured resumes and job descriptions into structured, searchable, and semantically comparable representations. The objective is not to build a complete HRMS; it is to develop and evaluate the resume-intelligence, job-intelligence, retrieval, and candidate-ranking modules that support recruiter decision-making.

- Resumes arrive in heterogeneous layouts, file types, and writing styles.
- Equivalent skills and responsibilities may be expressed using different terminology.
- Keyword-only systems provide limited recall and poor understanding of context.
- Large resume repositories require indexing and multi-stage retrieval rather than repeated file scanning.
- Recruiters require score breakdowns and evidence, not an unexplained similarity value.

1.1 Scope at the Mid-Semester Stage

The first three planned phases have been treated as completed for this report: Dissertation Outline and Setup, Design and Baseline Development, and Pipeline Development and Mid-Semester. The implementation currently covers resume upload and validation, text extraction, structured parsing, normalization, JD requirement extraction, baseline matching, semantic representations, candidate shortlisting, and initial ranking.

2. OBJECTIVES AND RESEARCH QUESTIONS

- Design a robust resume ingestion and parsing pipeline for PDF, DOC, and DOCX files.
- Extract structured candidate attributes such as skills, education, employment, roles, companies, certifications, and experience.
- Extract and classify JD requirements, including mandatory and preferred skills.
- Normalize candidate and job attributes into canonical representations.
- Compare keyword-based, TF-IDF, semantic, and hybrid candidate-retrieval approaches.
- Rank shortlisted candidates using structured fit and semantic evidence.
- Produce evidence-grounded recruiter-facing explanations.
- Evaluate extraction accuracy, retrieval relevance, ranking quality, latency, and scalability.

2.1 Research Questions

1. How accurately can a hybrid NLP pipeline extract ATS-relevant entities from heterogeneous resumes?
2. Does canonical skill and title normalization improve candidate retrieval?
3. Does hybrid lexical-semantic retrieval outperform keyword-only matching?
4. Does second-stage reranking improve Precision@K, Recall@K, MRR, and NDCG@K?
5. Can the ranking be explained using verifiable resume evidence?
6. What latency and storage trade-offs arise when the architecture is scaled toward one million candidate profiles?

3. SYSTEM MODULES COMPLETED UP TO MID-SEMESTER

3.1 Resume Upload and Validation

Accepts PDF, DOC, and DOCX resumes. The module checks file extension, MIME type, file size, corruption, duplicate file hash, and basic metadata before assigning a resume identifier and processing status.

3.2 File Storage and Resume Metadata

Stores the original resume through a file-storage abstraction. The database stores the storage path, original filename, content type, hash, size, upload time, version, candidate identifier, and processing state.

3.3 Text Extraction and OCR Routing

Extracts text using PDF and Word parsers. It calculates text-quality indicators and supports an OCR fallback for scanned or low-quality documents.

3.4 AI-Based Resume Parsing

Uses a hybrid combination of regular expressions, rules, skill dictionaries, entity recognition, and semantic methods to identify contact fields, skills, work history, education, certifications, title, company, projects, and experience.

3.5 Schema Validation and Quality Assessment

Validates parsed output against the ATS schema and records field confidence, missing attributes, warnings, provenance, parse status, and manual-review requirements.

3.6 Entity Normalization and Candidate Profile Creation

Maps raw values to canonical titles and skills, calculates total experience, and constructs a normalized candidate profile containing raw, extracted, normalized, provenance, quality, and embedding metadata.

3.7 Job Description Parsing

Extracts required and preferred skills, title, experience, education, certification, domain, responsibility, location, and employment type from a JD.

3.8 Candidate Indexing and Retrieval

Creates structured searchable fields, lexical representations, and semantic embeddings. Candidate retrieval uses indexed profile data instead of reopening all resume files.

3.9 Candidate Matching and Ranking

Combines required-skill coverage, preferred-skill coverage, title similarity, experience fit, domain fit, education, certification, and semantic similarity to rank the candidate shortlist.

3.10 Explainability Output

Produces candidate score breakdowns, matched skills, missing or unverified skills, experience fit, and evidence text for recruiter review.

4. FUNCTIONAL ARCHITECTURE AND PROCESSING FLOW

The functional architecture is divided into ingestion, document intelligence, normalization, indexing, JD intelligence, retrieval, reranking, and explanation layers. Figure 1 presents the processing flow implemented up to the mid-semester stage.

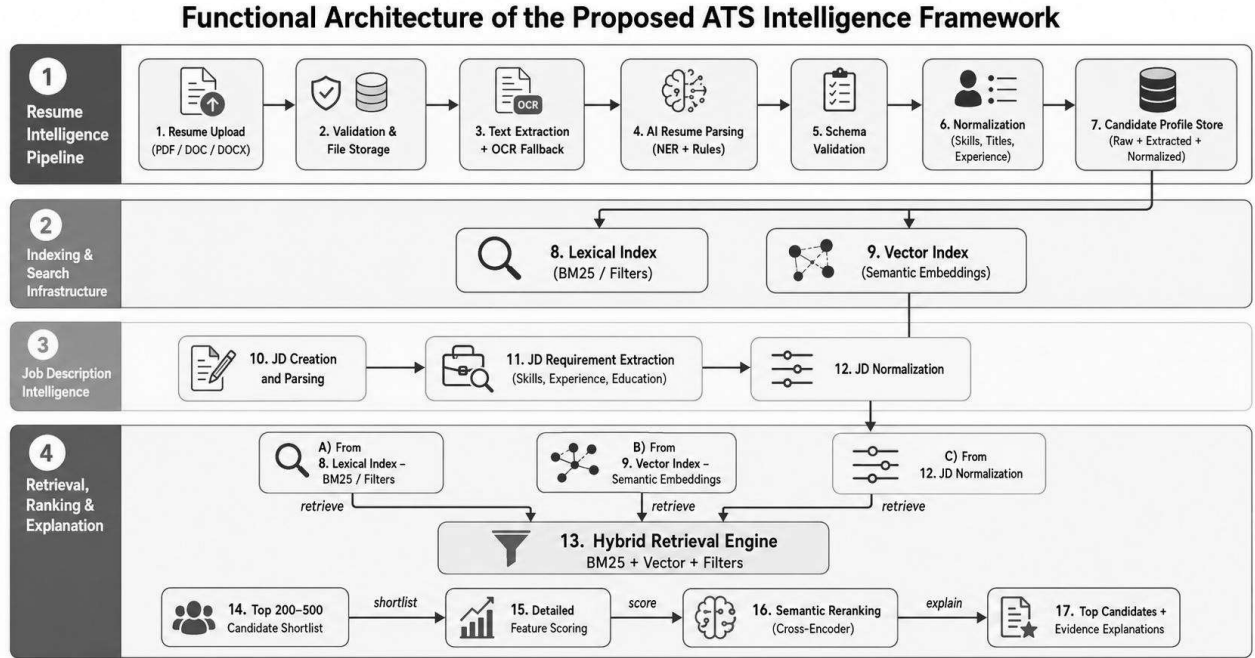


Figure: End-to-end pipeline for resume processing, job understanding, hybrid candidate retrieval, and explainable ranking.

Figure 1: Functional architecture of the proposed ATS intelligence framework

4.1 Resume Processing Flow

1. Upload and validate the resume.
2. Store the original document and metadata.
3. Extract raw text and use OCR only when required.
4. Parse ATS-relevant fields and retain evidence spans.
5. Validate the output schema and calculate quality indicators.
6. Normalize entities and construct the candidate profile.
7. Create structured, lexical, and semantic indexes.
8. Mark the candidate profile as searchable.

4.2 JD-to-Candidate Retrieval Flow

1. Create or upload a JD.
2. Extract and normalize mandatory and preferred requirements.
3. Apply only genuine hard filters.
4. Run BM25, skill/title filtering, and semantic retrieval in parallel.
5. Merge and deduplicate candidate identifiers.
6. Generate a shortlist of approximately 200-500 candidates.

7. Calculate detailed fit features and rerank the shortlist.
8. Return the top candidates with score breakdown and evidence.

5. DATA MODEL, STORAGE AND INDEXING DESIGN

5.1 Candidate and Resume Representation

The candidate is treated as the searchable entity, while each resume is maintained as a versioned source document. The normalized candidate profile consolidates the active resume and retains provenance back to source text.

Data Group	Representative Fields	Purpose
File metadata	filename, path, hash, MIME type, version	Storage, deduplication, audit
Raw	extracted text, page count, extraction method	Reprocessing and traceability
Extracted	skills, roles, employers, dates, education	Model output before normalization
Normalized	canonical skills, title, experience, seniority	Search and matching
Provenance	source span, section, page, method	Evidence and debugging
Quality	confidence, completeness, warnings, review flag	Human-in-the-loop control
Embeddings metadata	model, version, vector type, timestamp	Index governance and migration

5.2 Indexing Strategy

- Structured indexes: title, skills, total experience, education, location, certification, and status.
- Lexical index: normalized searchable text used by BM25 or full-text search.
- Semantic index: candidate-level and section-level embeddings for approximate nearest-neighbour retrieval.
- Only indexed profiles are searched when a JD is available; original files are reopened only for viewing, reprocessing, or evidence verification.

6. METHODOLOGY AND MODELS IMPLEMENTED

6.1 Information Extraction Methodology

- Regular expressions for email, phone, dates, and predictable patterns.
- Dictionary and taxonomy matching for skills, degrees, certifications, and abbreviations.
- Named Entity Recognition for organizations, roles, institutions, and contextual entities.
- Date normalization and employment-interval calculation for total experience.
- Confidence and provenance attached to each extracted entity.

6.2 Baseline Matching

- Exact and normalized keyword overlap.

- Weighted mandatory and preferred skill coverage.
- TF-IDF vectorization with cosine similarity.
- BM25-style lexical relevance for candidate retrieval.

6.3 Semantic Retrieval and Ranking

- Sentence embeddings for JDs, candidate summaries, experience, skills, and projects.
- Vector similarity for semantic candidate recall.
- Hybrid fusion of lexical rank, semantic rank, and structured match features.
- Initial weighted reranking using skill, title, experience, domain, education, and certification scores.
- Cross-encoder or learning-to-rank extension planned for final evaluation.

7. EXPERIMENTAL SETUP AND BASELINES

The evaluation is designed as a controlled comparison rather than a single-model demonstration. Public, synthetic, or consented resumes and representative JDs will be divided into development and evaluation sets. Candidate-JD relevance judgments will use graded labels such as highly relevant, relevant, partially relevant, and not relevant.

Experiment	Method	Primary Metrics	Status
E1	Rule/dictionary extraction	Precision, Recall, F1	Completed baseline
E2	Hybrid extraction	Field accuracy, entity F1	Initial pipeline completed
E3	Keyword overlap	Precision@K, Recall@K	Completed baseline
E4	TF-IDF similarity	MRR, NDCG@K	Completed baseline
E5	Dense semantic retrieval	Recall@K, latency	Initial implementation
E6	Hybrid retrieval	Recall@K, NDCG@K	Initial implementation
E7	Second-stage reranking	MRR, NDCG@K	Planned after mid-sem
E8	Ablation and scalability	Quality, p95 latency, storage	Planned after mid-sem

8. CURRENT IMPLEMENTATION STATUS

Work Package	Deliverable	Status	Evidence Expected
Literature and requirement study	Research matrix and scope	COMPLETED	Literature notes and architecture decisions
Project setup and datasets	Repository, environment, sample corpus	COMPLETED	Runnable environment and dataset manifest
System and schema design	Candidate, resume, JD, match schemas	COMPLETED	Schema definitions and API contracts
Baseline preprocessing	Cleaning and section handling	COMPLETED	Preprocessed sample outputs
Keyword/TF-IDF baseline	Initial candidate ranking	COMPLETED	Baseline metrics and ranked results
Resume extraction pipeline	Structured candidate fields	COMPLETED - INITIAL	Parsed JSON with confidence and provenance
JD intelligence pipeline	Normalized JD requirements	COMPLETED - INITIAL	Required/preferred skill outputs

Work Package	Deliverable	Status	Evidence Expected
Semantic representation	Embeddings and vector retrieval	COMPLETED - INITIAL	Vector index and retrieval results
Hybrid shortlisting	Lexical + semantic candidate pool	COMPLETED - INITIAL	Top candidate shortlist
Detailed ranking	Feature score and ranking	COMPLETED - INITIAL	Score breakdown per candidate
Full evaluation and ablation	Controlled comparison	PENDING	Final metric tables and analysis
User evaluation and final report	Recruiter review and dissertation	PENDING	Feedback, report, and presentation

9. MAJOR TECHNICAL SPECIFICATIONS OF THE PROPOSED ATS FRAMEWORK

Sl. No.	Technical Parameter	Proposed Specification
1	Supported resume formats	PDF, DOC, DOCX; image/scanned document path through OCR
2	Backend	Python-based API and asynchronous processing workers
3	Operational database	MongoDB-compatible flexible document model
4	Original document storage	Local storage for prototype; object-storage abstraction for scale
5	Resume parsing	Rules, regex, dictionaries, NER/transformer extraction, optional LLM comparison
6	Normalization	Canonical skills, titles, degrees, companies, dates, and experience buckets
7	Lexical search	Keyword/full-text or BM25 retrieval with field weighting
8	Semantic search	Sentence embeddings and vector similarity
9	Hybrid retrieval	Structured filters + lexical retrieval + semantic retrieval + rank fusion
10	Reranking	Weighted feature model initially; cross-encoder/LTR planned
11	Candidate output	Top-N candidates with component scores and evidence
12	Evaluation metrics	Precision, Recall, F1, Precision@K, Recall@K, MRR, MAP, NDCG@K
13	Scalability target	Architecture suitable for indexed retrieval over more than one million profiles
14	Security considerations	PII separation, access control, validation, auditability
15	Reproducibility	Versioned datasets, models, embeddings, indexes, configuration, and experiments

10. DESIGN CONSIDERATIONS

- Process each resume once and reuse stored structured and indexed representations.
- Separate original files, operational data, search indexes, and model artifacts.
- Treat the candidate as the searchable entity and resumes as versioned evidence sources.
- Do not reject a candidate solely because a skill was not found; represent the state as matched, not matched, or unverified.
- Use deterministic parsing for predictable fields and model-based extraction for contextual fields.
- Use a common normalization layer for resumes and JDs.
- Apply expensive reranking only to a retrieved shortlist.
- Retain provenance and confidence for explainability and manual correction.
- Exclude protected personal attributes from candidate ranking.
- Keep model and embedding versions to support controlled reprocessing and index migration.
- Design APIs and data structures for future multi-tenant ATS product integration.
- Validate research claims through baselines, ablations, and quantitative metrics.

11. PRELIMINARY OBSERVATIONS AND RISKS

11.1 Preliminary Observations

- Normalization is necessary because raw keyword overlap misses aliases such as ML and Machine Learning.
- Whole-document similarity can dilute important experience evidence; section-aware representations are therefore included in the design.
- Exact skill filters improve precision but can reduce recall when resumes use indirect terminology.
- Hybrid retrieval is more suitable than a single retrieval method because technical terms and semantic responsibilities behave differently.
- Reranking should operate on a limited shortlist to control latency and cost.

11.2 Current Risks and Mitigations

Risk	Mitigation
Limited labelled candidate-JD relevance data	Use graded human judgments, public benchmarks, and controlled synthetic cases.
OCR and complex-layout errors	Route poor-quality files through OCR/layout-aware processing and manual review.
Taxonomy coverage gaps	Use public taxonomies with a project-specific alias extension.
Embedding model bias or domain mismatch	Compare multiple models and perform subgroup/error analysis.
Scalability cost	Use multi-stage retrieval, bounded vectors per candidate, and synthetic load tests.
Unfaithful explanations	Generate explanations from stored evidence and score contributions.

12. FUTURE PLAN

The plan below follows the approved dissertation schedule. The first three phases are considered completed for the mid-semester report; the remaining phases focus on evaluation, refinement, review, and submission.

Sl. No.	Phase	Start Date - End Date	Work to be Done	Status
1	Dissertation Outline & Setup	25 Apr 2026 - 10 May 2026	Review literature, set up the project environment, and identify suitable resume and job description datasets.	COMPLETED
2	Design & Baseline Development	11 May 2026 - 31 May 2026	Design the framework and develop baseline preprocessing and keyword-based matching modules.	COMPLETED
3	Pipeline Development & Mid-Sem	01 Jun 2026 - 21 Jun 2026	Build resume extraction, job description understanding, semantic matching, and candidate ranking modules.	COMPLETED
4	Testing & User Evaluation	22 Jun 2026 - 12 Jul 2026	Evaluate extraction quality, matching accuracy, ranking relevance, conduct ablation studies, validate scalability, and compare results with baselines.	PENDING
5	Final Review & Submission	13 Jul 2026 - 25 Jul 2026	Incorporate feedback, finalize dissertation chapters, prepare presentation slides, conduct plagiarism checks, and complete final submission.	PENDING

13. ABBREVIATIONS

Abbreviation	Expansion
AI	Artificial Intelligence
ANN	Approximate Nearest Neighbour
API	Application Programming Interface
ATS	Applicant Tracking System

Abbreviation	Expansion
BM25	Best Matching 25
DOCX	Office Open XML Document
IE	Information Extraction
JD	Job Description
LTR	Learning to Rank
MRR	Mean Reciprocal Rank
NER	Named Entity Recognition
NLP	Natural Language Processing
OCR	Optical Character Recognition
PII	Personally Identifiable Information
RRF	Reciprocal Rank Fusion
SBERT	Sentence-BERT
TF-IDF	Term Frequency-Inverse Document Frequency
UI	User Interface

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